Theorem 5.8

Let S be a set. Then $S \subseteq S$ and $\emptyset \subseteq S$.

Proof.

Let S be an arbitrary set. By **Def 5.6**, we can show that a set S' is a subset of S by using the following the method to build S'. Let S' originally be the $\emptyset$. Then add to S', some elements also in S. Now, we can build S' in the following ways.

Add no elements of S into S'. Now $S' = \emptyset$. Thus, $\emptyset \subseteq S$.

Add all elements of S into S'. Now $S' = S$. Thus, $S \subseteq S$.

Therefore, if S is a set, then $S \subseteq S$ and $\emptyset \subseteq S$.

■

Theorem 5.18

Let A and B be sets. If $A \subseteq B$, then $B^c \subseteq A^c$.

Assume we have two sets, A and B. We then use proof by contrapositive. Thus we assume $B^c \nsubseteq A^c$. By Definition **5.6**, there exists $x \in B^c$ such that $x \notin A^c$. Definition **5.14** then tells us that x is in A but not in B. Hence, $A \nsubseteq B$. Thus, if $B^c \nsubseteq A^c$, then $A \nsubseteq B$. Therefore, Theorem **3.34** implies if $A \subseteq B$, then $B^c \subseteq A^c$.

■